Hunting mast cell progenitors in normal canine blood

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Abstract

Introduction. Mast cells are widely considered to be a multi-functional master cell, with involvement in histamine based allergic reactions, wound healing, tissue remodeling and innate immunity. They are strongly associated with the tumor micro environment as well as pathologies such as anaphylaxis, allergic rhinitis and infantile asthma as well as a myriad of other pathologies. Mast cells are found in nearly every tissue of the body except the central nervous system and the retina of the eye, but originate in the bone marrow as hematopoietic stem cells. After release from the bone marrow, they circulate in the blood as uncommitted mast cell progenitors (MCp) which are CD117⁺, CD34⁺, FcεRI⁺ and CD90⁻ before being recruited to the peripheral tissues where they mature. Malignant transformation of mast cells results in mast cell tumor, which is a common tumor of dogs. Canine mast cell tumors are most commonly found in the skin, but may also occur in the liver, intestines, spleen and elsewhere. Normal mast cells from canine mast cell tumor patients are needed to compare transcriptomes to patient mast cell tumors using deep sequencing/single cell sequencing and from this data, therapeutic targets may be selected. The current project seeks to identify and obtain appropriate normal mast cell precursors from patient animals with minimal additional morbidity. Circulating MCp's represent a population of approximately 0.1-0.5% of a cell types throughout the blood, making them a difficult target to isolate.

Methods. Herein we describe two methods for isolating a CD117⁺ cell population from whole blood. The first is a method of magnetic activated cell sorting (MACS) to deplete CD90⁺ cells and enrich for CD117⁺ cells followed by flow cytometric analysis. The second uses MACS to deplete CD90⁺ cells first followed by fluorescence activated cell sorting (FACS) to isolate CD117⁺ cells for downstream use in single cell sequencing.

Results. For the first method we start with roughly 3x10^6 cells and finish with roughly 5x10^4 cells. Cytospin has shown this to be a very heterogeneous population consisting of very few granulated cells. Flow cytometric analysis has also shown this to be a very heterogeneous population with only 1-2% CD117⁺ cells. Method two using FACS yields a much smaller cell population of about 8,000 cells.

Conclusions. Method 1 yields a very impure population that cannot be used for further downstream analysis via single cell sequencing. Method 2 yields a very low cell number but may be of higher purity so single cell sequencing may still be an option. We are currently troubleshooting both methods to increase purity enough for downstream usage in single cell sequencing.

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Method 1

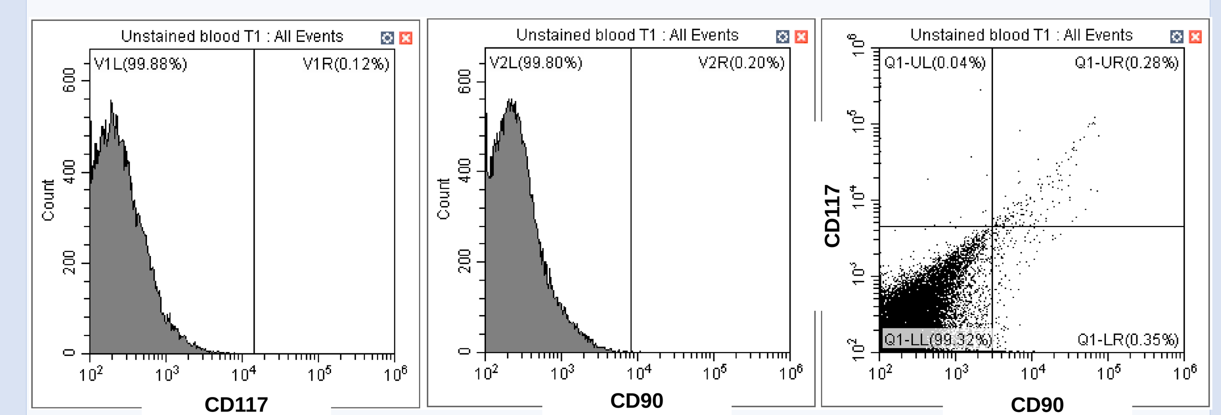
Isolate PBMCs
↓
Lyse RBCs
↓
Count Cells
↓
Block Cells
↓
Stain With Antibodies
↓
Magnetic Depletion for CD90
↓
Magnetic Enrichment for CD117
↓
Flow Analysis or Cytospin

Method 2

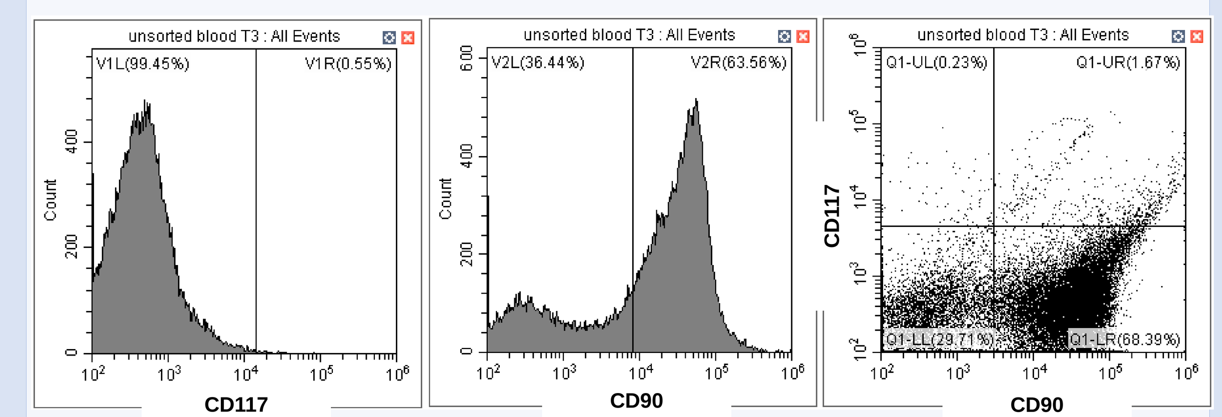
Isolate PBMCs
↓
Lyse RBCs
↓
Count Cells
↓
Block Cells
↓
Stain With Antibodies
↓
Magnetic Depletion for CD90
↓
Fluorescence Activated Cell Sorting
↓
Flow Analysis

Method 1 Results

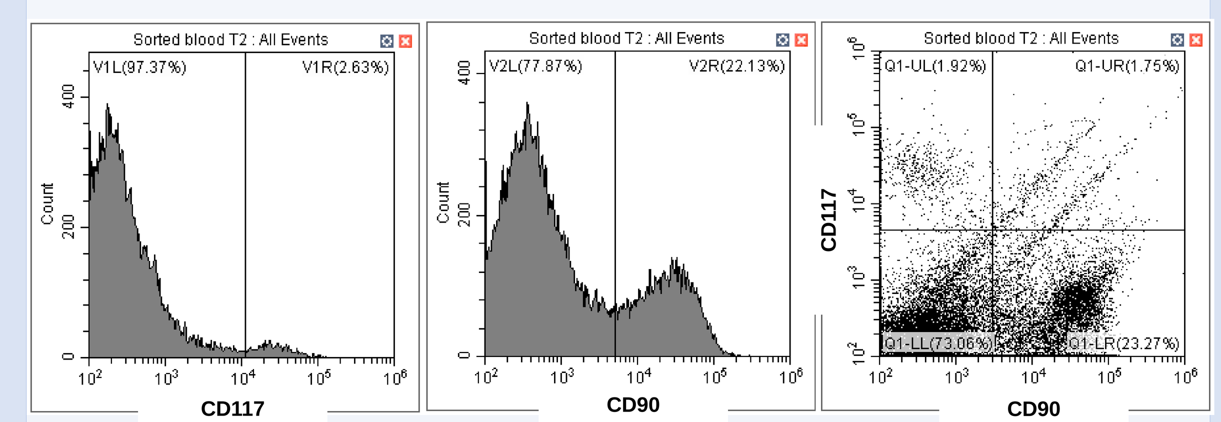
Unstained Sample



No Magnetic Sorting

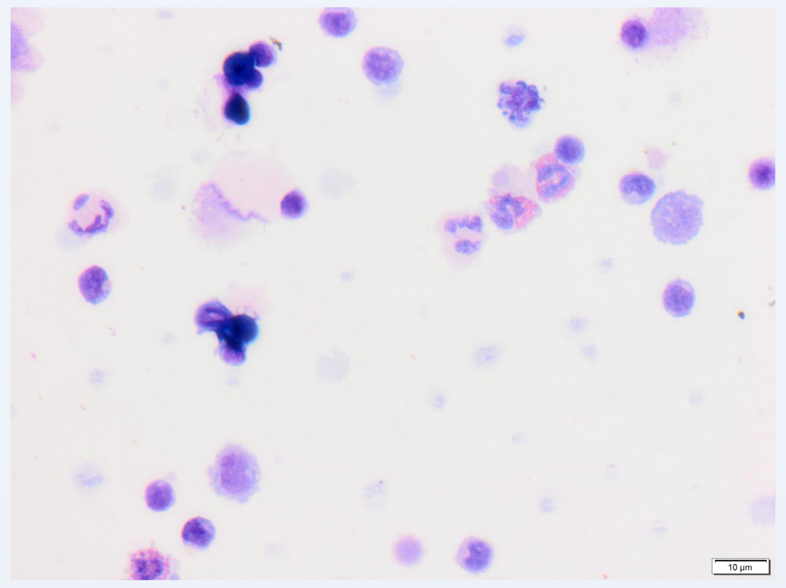


Magnetically Sorted

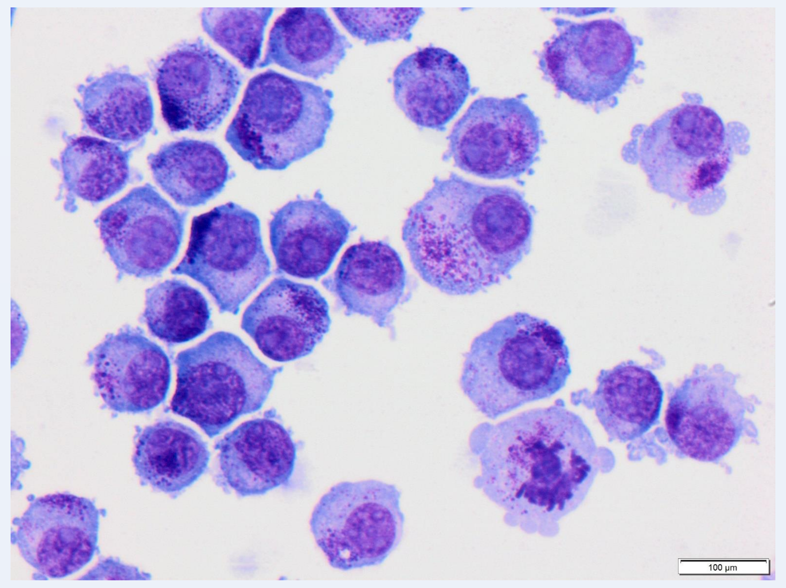


Cytospin Wright's Giemsa Stain Results

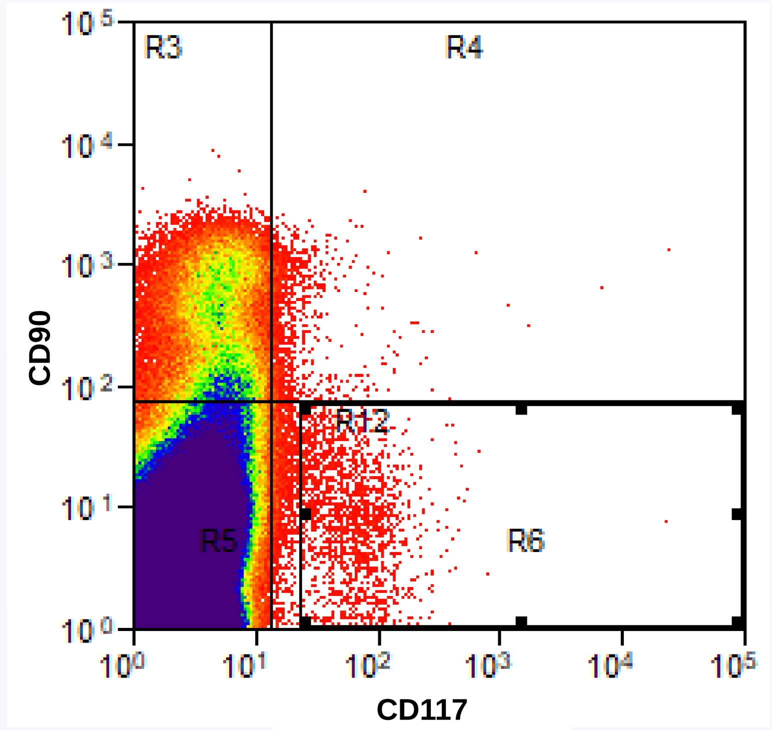
Blood

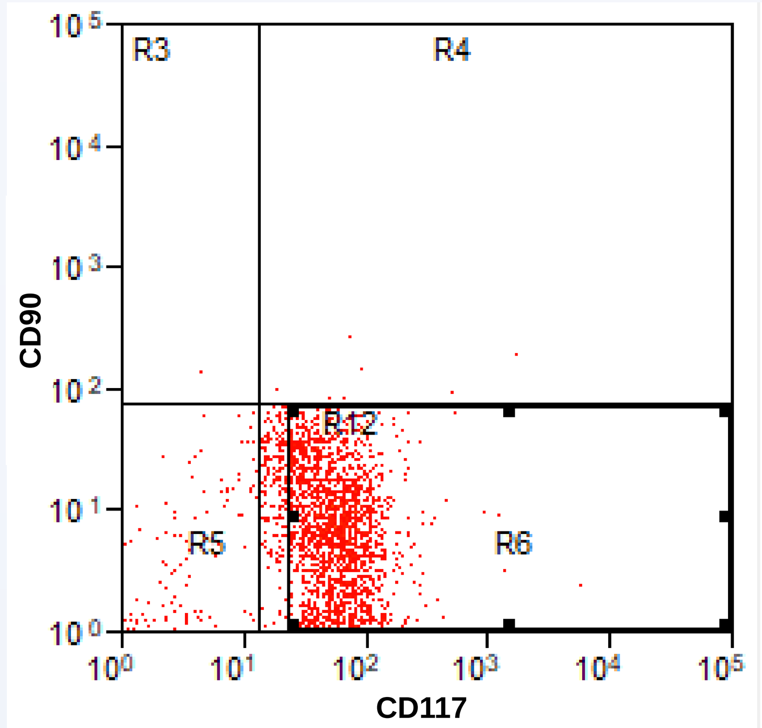


MPT-1 Mast Cells



Method 2 Results





Conclusions and Future Directions

- Method two works. We are able to obtain a pure CD117 positive cell population via fluorescence activated cell sorting.
- Method one only generates a 1.92% pure CD117 positive cell population, which is not sufficient for downstream application in single cell sequencing.
- Moving forward, we will be able to use method two to isolate CD117 positive cells for single cell sequencing. This cell population will contain our mast cell progenitors that may be identified by genetic markers and used as a normal comparison to mast cell tumor for patient treatment